

Software Requirement specification

Project Title: Farming Resource Sharing System

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Purpose:

The Farming Resource Sharing System is a collaborative platform designed to help farmers with diverse expertise—such as machinery operations, organic farming, and mixed cropping—work together effectively. The system streamlines collaboration by aligning farmers' skills, interests, and resources to maximize land productivity and achieve higher agricultural outcomes. It also provides a centralized platform for communication, resource management, and coordinated decision-making, enabling more efficient and sustainable farming practices.

Hardware Requirements:

Component	Specification
Processor	Intel Core i5 or higher
RAM	Minimum 8 GB
Storage	Minimum 20 GB free space
Operating System	Windows 10 / macOS / Linux (Ubuntu recommended)

Network	A stable internet connection is necessary to enable secure database API communication and user authentication through Clerk.
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Software Requirements:

Software	Version / Details
Frontend Framework	CSS, JavaScript , REACT ,
Backend Framework	Node.js , Express
Database	Neon db (postgreSQL)
IDE / Code Editor	Visual Studio Code (VS Code)
Version Control	Git and GitHub
Package Manager	Vite (Frontend Package Manager) / npm (Node Package Manager)
Browser	Chrome / Brave (latest version)

Functional Requirements:

Feature Area	Details and Functionality
User Authentication and Profiles	Allows users to register, log in, and create profiles containing details like name, phone number , email address
Group joining and Description	The farmer can view all available farming groups listed on the platform. Each group displays a brief summary including the type of resources involved. Based on this the farmer can choose the group that best matches their interests.
Farmers and Land display	Details and contact information of all registered farmers are displayed on the platform to support collaboration. Farmers can also add information about the land they own, including soil type, land area, and location.
Equipment lending and managing	The equipment management page allows farmers to list the machinery and tools they own and mark them as available for sharing. Farmers can view requests from others who need the equipment and also get required tools
Crops logging Feature	The farmer can enter the crops being cultivated along with their expected yield and current stage of growth.